

COMS 3003A

Test 2 – Deferred version

7 May, 2024

This is a closed book test.
Time allowance: 60 minutes

Caution: if, in any of your answers, you define a Turing machine and claim that it solves a particular problem, you have to prove your claim.

1. **(20 points)** Prove or disprove the following statements (i.e., if you believe the statement to be true, prove it; otherwise, provide a counterexample):
 - (a) **(5 points)** If languages L_1 and L_2 are undecidable, then $L_1 \cup L_2$ is undecidable.
 - (b) **(5 points)** Every countably infinite language is undecidable.
 - (c) **(5 points)** Every undecidable language is countably infinite.
 - (d) **(5 points)** If languages L_1 and L_2 are undecidable, then $L_1 \cap L_2$ is undecidable.
2. **(10 points)** Prove that the problem of determining whether the language of a Turing machine contains exactly 4 words is algorithmically unsolvable, i.e., that the language $\{\langle M \rangle : |L(M)| = 4\}$ is undecidable. You may assume that the following problems are unsolvable:
 - non-self-acceptance for Turing machines;
 - self-acceptance for Turing machines;
 - acceptance for Turing machines.
3. **(20 points)** Does there exist a bijection between the set of all deciders and the set of all undecidable decision problems? Prove your statement. Your proof **may not** assume any fact proven in class or in textbook—if you need to use a fact in your proof, you need to prove it first.
4. **(10 points)** The first-order language of set theory contains a binary predicate symbol $\in$. Translate the following statements into this language (you are free to define any other symbols in terms of the ones listed above):
 - (a) **(5 points)** There exists a set that contains exactly one element.
 - (b) **(5 points)** For every pair of sets X and Y , there exists a set whose elements are the common elements of X and Y .